<epam>

Business Template

RECRUITMENT AGENCY



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1. BUSINESS DESCRIPTION

1.1. BUSINESS BACKGROUND

In today's competitive job market, connecting qualified candidates with potential employers is essential. Recruitment agencies must efficiently manage candidate profiles, job listings, applications, and interview processes to remain competitive. The goal of this project is to develop a robust and efficient database model that will streamline these processes.

1.2. PROBLEMS. CURRENT SITUATION

Currently, many recruitment and recruitment agencies struggle with fragmented data management systems. Data such as candidate profiles, job requirements, interview schedules, and training services are often stored in separate systems or even manually maintained. This leads to inconsistencies, data loss, and inefficiencies in the hiring process. Additionally, tracking candidate skills and matching them with job requirements can be challenging without a unified system.

1.3. THE BENEFITS OF IMPLEMENTING A DATABASE. PROJECT VISION

The primary benefit of the proposed database is to centralize and streamline data management for recruitment and recruitment processes. By integrating candidate profiles, job listings, applications, interviews, and skill development services into a single database, the system will enable efficient data access and manipulation. This will reduce the time needed for data retrieval, improve matching accuracy between candidates and job requirements, and enhance overall decision-making efficiency.

2. MODEL DESCRIPTION

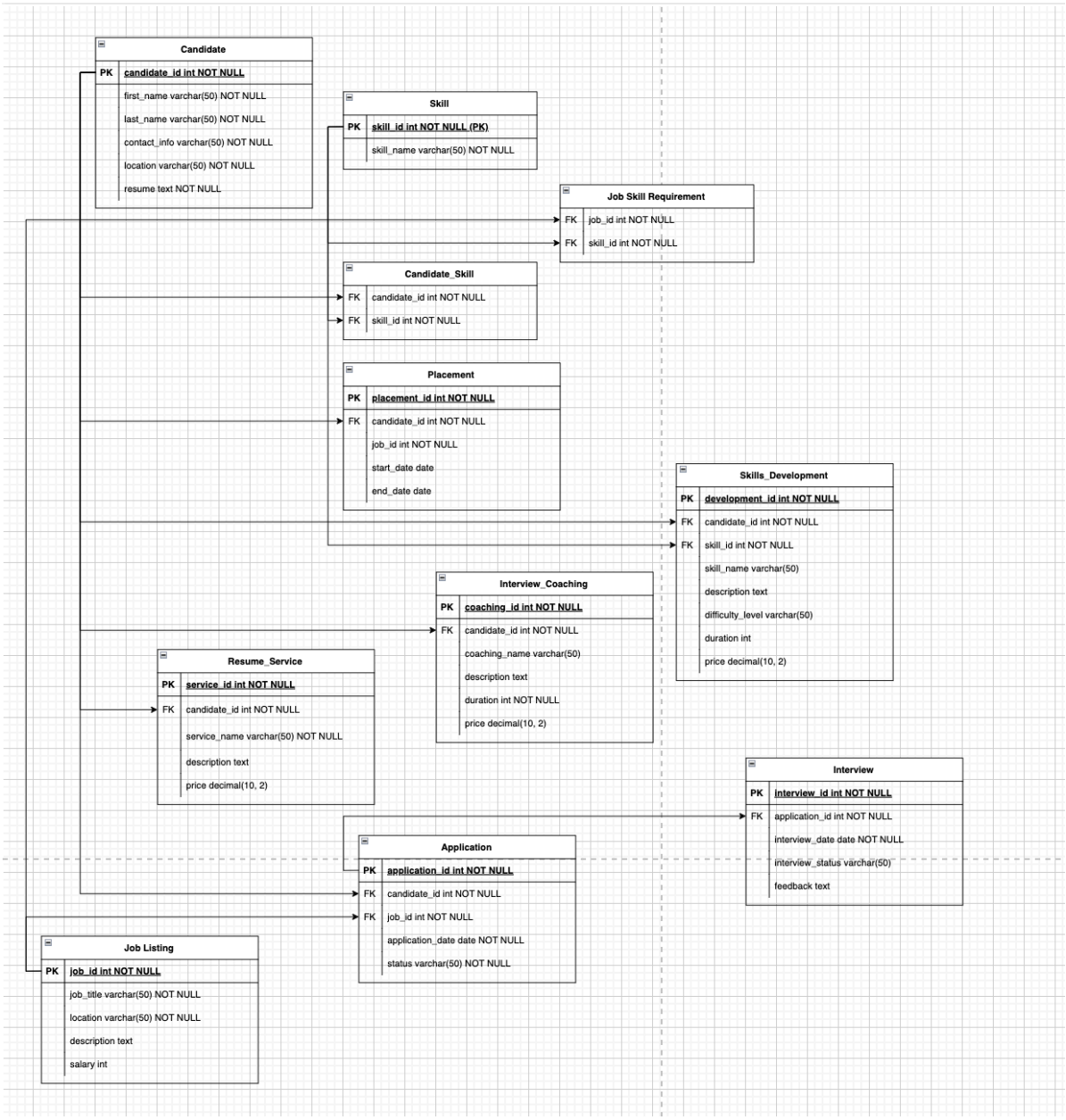
2.1. DEFINITIONS & ACRONYMS

- PK: Primary Key
- FK: Foreign Key
- 3NF: Third Normal Form
- M:N: Many-to-Many Relationship

2.2. LOGICAL SCHEME

The database model follows the principles of the Third Normal Form (3NF), ensuring minimal redundancy and maintaining data integrity. The model consists of 11 tables, covering all essential aspects of the recruitment process, including candidate management, job listings, applications, interviews, and skill development.

The model includes a many-to-many relationship between Candidates and Skills, which is managed through a junction table called Candidate_Skill. Additionally, the relationship between Jobs and Skills is established through the Job_Skill_Requirements table.



2.3. OBJECTS

The model includes the following tables:

1. **Candidate**: Stores candidate information such as name, contact details, and resume.

Table Name	Field name	Field Description	Data Type
Candidate	candidate_id	Unique identifier for each candidate, PK	Int
	first_name	candidate's first name	varchar(50)
	contact_info	candidate's contact information	varchar(50)
	resume	Text field containing the candidate's resume.	Text
	Location	Geographical location of the candidate.	varchar(50)

2. **Skill**: Stores available skills.

Table Name	Field name	Field Description	Data Type
Skill	skill_id	Unique identifier for each skill, PK	Int
	skill_name	Name of the skill (e.g., Python, Data Analysis).	varchar(100)

3. **Candidate_Skill**: Represents the many-to-many relationship between candidates and skills.

Table Name	Field name	Field Description	Data Type
Candidate_skill	candidate_id	Reference to the candidate possessing the skill, FK	Int
	skill_id	Reference to the skill the candidate has, FK	Int

4. **Job Listing**: Contains job titles, descriptions, locations, and salaries.

Table Name	Field name	Field Description	Data Type
Job Listing	job_id	Unique identifier for each job listing, PK	Int
	job_title	Title of the job position (e.g., Software Developer).	varchar(50)
	location	Location where the job is based.	varchar(50)
	description	Detailed description of the job role and responsibilities.	Text
	salary	Expected salary for the position.	Int

5. **Job Skill Requirement**: Specifies required skills for a job.

Table Name	Field name	Field Description	Data Type
Job Skill Requirement	job_id	Reference to the job listing requiring a specific skill, FK	Int
	skill_id	Reference to the required skill for the job, FK	Int

6. **Application**: Stores applications submitted by candidates for specific job listings.

Table Name	Field name	Field Description	Data Type
Application	application_id	Unique identifier for each application, PK	Int
	candidate_id	Reference to the candidate submitting the application, FK	Int
	job_id	Reference to the job being applied for, FK	Int
	application_date	Date when the application was submitted.	Date
	Status	Current status of the application (e.g., Pending, Accepted).	varchar(50)

7. **Interview**: Records interview details linked to applications.

Table Name	Field name	Field Description	Data Type
Interview	interview_id	Unique identifier for each interview, PK	Int
	application_id	Reference to the application being interviewed, FK	Int
	interview_date	Status of the interview (e.g., Scheduled, Completed).	Date
	interview_status	Status of the interview (e.g., Scheduled, Completed).	varchar(50)
	feedback	Feedback or notes from the interview.	Text

8. **Placement**: Tracks successful placements and contract dates.

Table Name	Field name	Field Description	Data Type
Placement	placement_id	Unique identifier for each placement record, PK	Int
	candidate_id	Reference to the candidate who was placed, FK	Int
	job_id	Reference to the job where the candidate was placed, FK	Int
	start_date	Start date of the employment.	Date
	end_date	End date of the employment (if applicable).	Date

9. **Resume_Service**: Stores details about resume enhancement services.

Table Name	Field name	Field Description	Data Type
Resume Service	service_id	Unique identifier for each resume service, PK	Int
	candidate_id	Reference to the candidate using the service, FK	Int
	service_name	Name of the resume service provided.	varchar(50)
	description	Description of the resume enhancement service.	Text
	price	Cost of the service.	decimal(10, 2)

10. **Interview_Coaching**: Offers coaching sessions to improve interview performance.

Table Name	Field name	Field Description	Data Type
Interview Coaching	coaching_id	Unique identifier for each coaching session, PK	Int
	candidate_id	Reference to the candidate receiving coaching, FK	Int
	coaching_name	Name of the coaching service.	varchar(50)
	description	Description of the coaching session.	Text
	duration	Duration of the session (in hours).	Int
	price	Cost of the coaching session.	decimal(10, 2)

11. **Skills_Development**: Tracks skill enhancement programs.

Table Name	Field name	Field Description	Data Type
Table 1	development_id	Unique identifier for each skills development program.	Int
	candidate_id	Reference to the candidate attending the program.	Text
	skill_id	Reference to the skill being developed.	
	skill_name	Name of the skill being developed.	
	description	Description of the training program.	
	difficulty_level	Difficulty level of the training (e.g., Beginner, Advanced).	
	duration	Duration of the program (in hours).	
	price	Cost of the training program.	